

Peptic Ulcer Disease Detection with Baseline Classifiers and LLM- Assisted Recommendation Generation for Care Support

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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the
Requirements for the **Degree of Bachelor of Science in
Computer Science and Engineering**

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Dhaka, Bangladesh

January 06, 2026

APPROVAL

This Project titled “**Peptic Ulcer Disease Detection with Baseline Classifiers and LLM-Assisted Recommendation Generation for Care Support**” submitted by **Md Murshedul Khoyar, ID: 213-15-4483** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **06-01-2026**.

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We hereby declare that this project has been done by me under the supervision of **Dr. Sheak Rashed Haider Noori, Professor & Head**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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ACKNOWLEDGEMENTS

This work would not have been possible without the support and contributions of many individuals over the past two semesters. We are deeply grateful to everyone who has assisted us in one way or another.

First, we express our heartfelt thanks and gratefulness to the almighty for His divine blessing making it possible for us to complete the **Final Year Design Project(FYDP)** successfully.

We are grateful and wish our profound indebtedness to **Dr. Sheak Rashed Haider Noori, Professor & Head**, Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh. Deep knowledge and keen interest of our supervisor in the field of **Large Language Model (LLM)** to carry out this project. His endless patience, scholarly guidance, continual encouragement, constant and energetic supervision, constructive criticism, valuable advice, reading many inferior drafts, and correcting them at all stages have made it possible to complete this project.

We would like to express our heartfelt gratitude to the Head of the Department of Computer Science and Engineering, for his kind help in finishing our project and also to other faculty members and the staff of the Department of Computer Science and Engineering, Daffodil International University.

We would like to thank our entire course-mates at Daffodil International University, who took part in this discussion while completing the coursework.

Finally, we must acknowledge with due respect the constant support and patience of our parents.

ABSTRACT

Accurate interpretation of upper gastrointestinal endoscopy reports is essential for the early detection and monitoring of peptic ulcer disease and related upper GI disorders. Manual reporting is often time-consuming and subjective, and simple rule-based or keyword-matching systems do not generalize well across different clinicians, hospitals or evolving documentation styles. To address this, we propose a robust clinical decision support pipeline that combines classical machine learning with modern language models to automatically analyze endoscopy findings and generate standardized diagnostic comments and advice. Using a real-world peptic ulcer dataset stored in Excel format, we first clean and preprocess the textual fields for the oesophagus, stomach and duodenum, along with existing free-text comments and recommendations. The text is normalized through lowercasing, removal of non-alphabetic characters, tokenization, stop-word removal and lemmatization, and then transformed into TF-IDF feature representations that capture the most informative clinical terms and phrases. On top of these features, separate Logistic Regression classifiers are trained to predict structured diagnostic comments and follow-up advice, effectively converting noisy free text into consistent labels. The best models are selected based on performance on held-out test data, using metrics such as accuracy, precision, recall and F1-score, showing reliable and clinically meaningful behaviour. To bridge the gap between machine outputs and human understanding, the predictions, together with patient demographics and original findings, are passed to a Large Language Model that generates clear, patient-friendly summaries and tailored lifestyle or follow-up suggestions. This design makes the system both interpretable for doctors and understandable for patients. Because the pipeline is lightweight and built from modular components, it can be periodically retrained or extended as new labeled cases become available, without redesigning the entire architecture. Overall, the combination of robust text preprocessing, TF-IDF-based classification and LLM-driven explanation provides an effective yet flexible decision support system for peptic ulcer management, with strong potential for integration into real-world clinical workflows where medical data and reporting patterns continuously evolve.

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Chapter 1

Introduction

We present some basic information about our work in this chapter, including the background of peptic ulcer disease, our motivation, project objectives, the chosen methodology, expected outcomes, and the organization of the report.

1.1 Introduction

Peptic ulcer disease (PUD) is a common upper gastrointestinal disorder where ulcers develop in the stomach or duodenum. In routine practice, upper GI endoscopy is performed to visualize these lesions and the endoscopist writes a report that includes patient information, the indication for the procedure, and narrative descriptions of the oesophagus, stomach and duodenum. At the end of the report, a short “Comment” summarizes the diagnostic impression, and an “Advice” section mentions the suggested management plan. In our setting these reports are stored in an Excel file, which contains patient ID, age, sex, visit date, indication, procedure, medication, findings in different GI segments, biopsy information, comments and advice. Although this dataset is rich, most of the important information is written as free text, which makes automatic analysis and consistent decision support difficult.

In this work we focus on this real endoscopy dataset and use a Python notebook to build a text-based decision support framework. Instead of analyzing endoscopic images, we work directly on the textual findings of the oesophagus, stomach and duodenum. We preprocess these texts, convert them into TF-IDF features and train baseline logistic regression models to predict the diagnostic comment (six classes) and the advice (five classes). On top of these models, we integrate a large language model (LLM) through an API, which receives patient details, endoscopy findings and the model predictions and then generates a more detailed narrative suggestion. In this way the system combines classical machine learning with modern LLM-based generation to support care for peptic ulcer patients.

1.2 Motivation

Peptic ulcer disease can cause serious complications such as bleeding and gastric outlet obstruction, so timely diagnosis and appropriate advice are very important. However, in busy hospitals most endoscopy reports are written in free text, and different doctors may describe similar findings and give similar advice in different ways. This variation makes it hard to reuse the data for research, quality monitoring or automated decision support. Reviewing more than a thousand reports manually is also time-consuming. At the same time, machine learning and natural language processing now offer practical

tools for working with such text data, and large language models can help generate clear explanations and recommendations.

We were therefore motivated to explore whether a hybrid text-based approach could be learned from the existing reports in our dataset and provide at least a basic level of automated support. As computer science students, this project also allows us to apply what we have learned about Python, data preprocessing, feature extraction, classification algorithms and API integration to a real medical problem. Our goal is not to replace doctors, but to design a prototype that shows how unstructured endoscopy text can be turned into structured labels and narrative recommendations.

1.3 Objectives

The main objective of this project is to develop a simple but complete AI-based framework that uses upper GI endoscopy report text to support peptic ulcer disease detection and care recommendation. More specifically, we aim to preprocess the peptic ulcer dataset, extract TF-IDF features from the oesophagus, stomach and duodenum descriptions, train baseline logistic regression models that predict the final diagnostic comment and advice, and then connect these predictions with a large language model that can generate a detailed, human-readable suggestion for care support based on a given patient record.

1.4 Methodology

First, we load the Excel dataset into a pandas DataFrame and perform basic cleaning such as converting the visit date into a proper date type and filling missing indication values with a default text. Then we apply a custom text preprocessing function on the oesophagus, stomach and duodenum columns, where we convert text to lowercase, remove non-alphabetic characters and normalize tokens. The cleaned text is transformed into numerical vectors using the TF-IDF scheme, and the resulting sparse matrices are combined into a single feature matrix.

Next, we encode the original “Comments” into six numeric classes and “Advice” into five numeric classes. We split the data into training and testing sets and trained two logistic regression models: one for predicting the diagnostic comment and another for predicting the advice. These models are evaluated on the test set using common classification metrics and confusion matrices. Finally, for a given patient record, we reuse the same preprocessing and feature extraction steps, predict the most likely comment and advice, and construct a structured prompt containing demographics, endoscopy findings and model outputs. This prompt is sent to an external large language model, which returns a detailed narrative recommendation that explains the likely diagnosis and suggests a management plan with a reminder that final decisions belong to the clinician.

1.5 Project Outcome

By following this approach, the project produces a cleaned and analyzed peptic ulcer dataset and a working Python pipeline for text preprocessing, TF-IDF feature extraction and baseline classification of comments and advice. We obtain quantitative results that show the performance and limitations of logistic regression when trying to reproduce clinicians’ written impressions from short text fields. In addition, we develop an LLM-based recommendation module that can generate longer narrative suggestions for real patient cases, using both the raw findings and the baseline model predictions. Overall, the outcome is a proof-of-concept system that demonstrates how existing upper GI endoscopy reports can be transformed into structured information and supportive

recommendations for peptic ulcer disease care.

1.6 Organization of the Report

This report is divided into several chapters. Chapter 1 presents the introduction, motivation, objectives, methodology and expected outcomes of the project. Chapter 2 discusses the necessary background and reviews related work on peptic ulcer disease, AI in gastrointestinal endoscopy, text analysis of clinical reports and the use of large language models in healthcare and then identifies the research gap. Chapter 3 describes the research methodology in more detail, including dataset description, preprocessing steps, feature extraction, model design and overall system architecture. Chapter 4 focuses on the implementation of the Python pipeline, the experimental setup and the results obtained from the logistic regression models and the LLM-assisted recommendation module. Chapter 5 summarizes the main findings, discusses engineering and ethical issues, and outlines .

Chapter 2

Background

In this chapter we describe the background of our work and the related literature. First, we give a short introduction of peptic ulcer disease and AI in gastrointestinal (GI) care. Then we discussed the important research papers we studied, mainly about image-based peptic ulcer detection, machine learning for GI bleeding risk prediction, and large language models (LLMs) in gastroenterology. Finally, we present a gap analysis table which shows how our proposed work is different.

2.1 Introduction

Peptic ulcer disease (PUD) is a common upper gastrointestinal disorder where ulcers develop in the stomach or duodenum. It is strongly associated with *Helicobacter pylori* infection, use of nonsteroidal anti-inflammatory drugs and other risk factors, and can lead to complications such as upper GI bleeding, perforation and gastric outlet obstruction if not treated properly. In modern hospitals, upper GI endoscopy is the standard procedure for visualizing these ulcers, documenting their location and severity, and deciding on management. In recent years researchers have tried to support this process with artificial intelligence. A large part of the literature uses deep learning on endoscopic images to detect ulcers, bleeding and other lesions [1]–[4], and another group of works uses machine learning on structured clinical data to predict outcomes like re-bleeding, need for intervention and mortality in GI bleeding [5]–[10]. At the same time, several review papers discuss how AI and especially LLMs may help in digestive endoscopy and gastroenterology by assisting decision making, documentation and patient education [11]–[15].

Most of these studies are very useful, but they mainly focus on either image-based lesion detection or numerical risk prediction. They do not show a complete system where a simple supervised model predicts peptic-ulcer-related labels from text or structured data and then an LLM converts those predictions plus patient information into a detailed, human-readable recommendation. Our project tries to fill this gap on top of our own peptic ulcer dataset.

2.2 Literature Review

In this section we describe the main research papers we reviewed in a combined way. Mahmood et al. proposed a robust deep learning model to classify peptic ulcer and other digestive tract disorders using endoscopic images and a stacked convolutional neural network with strong preprocessing and augmentation, and they reported high accuracy for ulcer and non-ulcer classes, which shows that image-based AI can be very powerful for lesion detection [1]. Yen et al. compared a convolutional neural

network with novice and experienced endoscopists for classification of bleeding peptic ulcer stigmata, and they found that the deep learning model could match or exceed novice performance for bleeding severity grading, which means AI can support less experienced doctors in emergency ulcer bleeding cases [2]. Singh et al. designed an AI-based peptic ulcer detection system using deep convolutional neural networks and reported good diagnostic accuracy in a diagnostic accuracy study [3]. Kuchkorov et al. used a lightweight MobileNetV3-Small architecture to detect gastric ulcers in endoscopic images and showed that even compact models can localize ulcers effectively, which is helpful for deployment on devices with limited resources [4]. Bajhaiya et al. then moved one step further and used a DenseNet121-based model to generate artificial endoscopic images of gastric ulcers, showing that deep learning can also produce realistic synthetic ulcer images for data augmentation and training [12].

On the structured-data side, Wong et al. developed a machine learning model to predict recurrent ulcer bleeding in patients with a history of idiopathic gastroduodenal ulcer bleeding, using clinical and endoscopic variables; their model can identify low-risk patients who may not need long-term proton pump inhibitor therapy, which shows how ML can change long-term treatment decisions in PUD [5]. Li et al. performed a bibliometric analysis of machine learning applications in acute upper GI bleeding and showed that research on this topic has grown quickly, with many studies focusing on outcomes such as mortality, transfusion and endoscopic therapy, but they did not propose a specific model themselves [6]. Nigam et al. wrote a narrative review summarizing machine learning models used for assessment and management of acute GI bleeding, and they concluded that ML often outperforms traditional clinical scores like Glasgow-Blatchford and Rockall for predicting mortality and need for intervention, but they also pointed out issues related to data quality and external validation [7]. Boros et al. built XGBoost and CatBoost models for mortality prediction in GI bleeding and found that these models performed better than conventional risk assessment tools, suggesting that tree-based ensemble methods are very promising for this problem [8]. Raghareutai et al. developed and validated a simplified ML model to predict which patients with upper GI bleeding need endoscopic intervention; their model can help primary care doctors decide whom to refer urgently for endoscopy, especially in settings where resources are limited [9]. Tsai et al. proposed machine learning models to predict mortality in patients with cirrhosis and upper GI bleeding, again showing that structured EHR data combined with ML can accurately identify high-risk patients at an early stage [10].

Alongside these task-specific models, several review papers discuss the broader use of AI and LLMs in digestive endoscopy and gastroenterology. Zhao et al. reviewed the application status and prospects of AI in digestive endoscopy and reported that convolutional neural networks achieve high sensitivity and specificity for detection of ulcers, polyps and other lesions, but they also mentioned challenges like dataset bias, lack of real-time prospective studies and the need for clinical validation before

routine use [11]. Wiest et al. discussed how large language models can be used for clinical decision support in gastroenterology and hepatology, including summarizing guidelines, answering complex clinical questions and guiding diagnosis and treatment steps, and they stressed that such models must be carefully controlled to avoid hallucinations and unsafe advice [13]. Gong et al. performed a systematic review of LLMs in gastroenterology and found early evidence that LLMs can help in endoscopy report generation, question answering and education on colorectal cancer and inflammatory bowel disease, but most studies are small and simulation-based instead of being integrated into real hospital systems [14]. Maida et al. also reviewed the application of large language models in gastroenterology and concluded that tools like ChatGPT have high potential for diagnostic support, education and literature interpretation, but that full integration into clinical workflows and safety validation is still missing [15].

From all these works we can see that deep learning on images is already strong for ulcer detection [1]–[4], [11], structured machine learning models are effective for GI-bleeding risk prediction and outcome estimation [5]–[10], and LLMs are starting to be used for documentation and decision support in gastroenterology [13]–[15]. At the same time, none of these studies combine a simple, transparent supervised model trained on a peptic-ulcer-focused dataset with an LLM that generates customized care recommendations based on the model prediction and patient information, which is the main direction of our work.

Table 2.1: Summary of Literature Reviewed.

Author (s)	Year	Title	Methodology	Key Findings
Mahmood et al.	2022	A Robust Deep Model for Classification of Peptic Ulcer and Other Digestive Tract Disorders Using Endoscopic Images	Large endoscopic image dataset; deep CNN with preprocessing and augmentation	Achieved high accuracy for ulcer and non-ulcer classification; shows strong potential of image-based AI for digestive disorders.
Yen et al.	2021	Performance Comparison of the Deep Learning and the Human Endoscopist for Bleeding Peptic Ulcer Disease	CNN trained on bleeding ulcer images; compared with novice and expert endoscopists	Deep learning model matched or exceeded novice performance in grading bleeding stigmata of peptic ulcers.
Singh <i>et al.</i>	2025	AI-Based Peptic Ulcer Detection Using Deep Convolutional Neural Networks: A	Deep CNN applied to endoscopic images in a diagnostic accuracy design	Reported good diagnostic accuracy for detecting ulcerative lesions; focuses on image-level classification.

		Diagnostic Accuracy Study		
Kuchkoro v et al.	2024	Gastric Ulcer Detection in Endoscopic Images Using MobileNetV3-Small	Lightweight MobileNetV3-Small network on gastric endoscopy images	Demonstrated effective ulcer detection with a compact model suitable for resource-limited devices.
Wong et al.	2019	Machine Learning Model to Predict Recurrent Ulcer Bleeding in Patients with History of Idiopathic Gastrointestinal Ulcer Bleeding	ML model trained on clinical and endoscopic features	Identified low-risk patients for recurrent bleeding, potentially reducing unnecessary long-term PPI therapy.
Li et al.	2024	Application of Machine Learning in Acute Upper Gastrointestinal Bleeding: Bibliometric Analysis	Bibliometric study of ML-related AUGIB publications	Showed rapid growth of ML research on AUGIB; mapped main topics and trends but no new model.
Nigam et al.	2024	Machine Learning in the Assessment and Management of Acute Gastrointestinal Bleeding	Narrative review of existing ML models and scores	Concluded ML often outperforms traditional scores for predicting adverse outcomes in acute GI bleeding.
Boros et al.	2025	New Machine-Learning Models Outperform Conventional Risk Assessment Tools in Gastrointestinal Bleeding	XGBoost and CatBoost models using structured clinical data	ML models outperformed conventional risk scores for mortality prediction in GI bleeding patients.
Raghareutai et al.	2025	Development and Validation of a Machine Learning Model for Predicting Endoscopic	Prospective UGIB dataset; simplified ML model for intervention prediction	Accurately identified patients needing urgent endoscopy; useful for triage in limited-resource settings.

		Intervention in Upper Gastrointestinal Bleeding		
Tsai et al.	2024	Machine Learning Models for Predicting Mortality in Patients with Cirrhosis and Upper Gastrointestinal Bleeding	Multiple ML algorithms on EHR data of cirrhotic patients	Achieved high performance for early mortality prediction in cirrhosis with UGIB.
Zhao et al.	2022	Application Status and Prospects of Artificial Intelligence in Digestive Endoscopy	Narrative review of AI systems in digestive endoscopy	Reported high sensitivity/specificity of CNNs for ulcers and other lesions; highlighted need for prospective validation.
Bajhaiya et al.	2023	Deep Learning-Powered Generation of Artificial Endoscopic Images of Gastric Ulcers	DenseNet121-based model for ulcer detection and synthetic image generation	Generated realistic artificial ulcer images; useful for data augmentation and training of detection models.
Wiest et al.	2025	Large Language Models for Clinical Decision Support in Gastroenterology and Hepatology	Perspective/review on LLM-based CDSS	Outlined potential uses of LLMs across GI pathways; emphasized risks like hallucination and need for safeguards.

2.3 Gap Analysis

Table 2.2: Gap Analysis

Features	Image-based Deep Learning Models	GI-bleeding ML Models (tabular data)	LLM-based Tools in Gastroenterology	Proposed System
Uses upper GI endoscopy report text (findings, comment, advice) as main input	NO	NO	SOMETIMES	YES
Works without endoscopy images (only text / structured fields)	NO	YES	YES	YES
Focuses on peptic ulcer disease with comment and advice structure	YES (ulcer only, image level)	PARTLY	NO	YES
Predicts both diagnostic impression and management advice	NO	NO	NO	YES
Integrates supervised ML predictions with an LLM in a single pipeline	NO	NO	NO	YES
Generates patient-specific narrative recommendation in simple language	NO	NO	SOMETIMES	YES
Designed for small single-centre Excel dataset (low-resource setting)	NO	SOMETIMES	NO	YES

2.4 Summary

In this chapter we have discussed the background and literature related to our work. We saw that many researchers have used deep convolutional neural networks to detect peptic ulcers and related lesions directly from endoscopic images, with high accuracy but also with high data and computation requirements. We also observed that several machine learning models built on structured clinical data can predict important outcomes in upper GI bleeding, such as recurrent bleeding, need for intervention and mortality, often performing better than traditional clinical scores. Finally, we reviewed recent discussions and systematic reviews on the role of large language models in gastroenterology, which show promising use cases in decision support, documentation and education, but are still in early stages of real clinical integration. From the gap analysis we can conclude that there is very little work that uses only the text and structured information of a peptic-ulcer-focused dataset to predict diagnostic impressions and management advice and then combines those predictions with an LLM to produce patient-specific recommendations. Our proposed system aims to fill this space using a baseline text-feature-based classifier and a recommendation-generating LLM on top of our own upper GI endoscopy dataset.

Chapter 3

Research Methodology

This chapter describes the complete process used to build our peptic ulcer disease detection and care-support system. It explains how the real endoscopy dataset was prepared, how text features were extracted, how baseline machine learning models were trained for diagnosis and advice prediction, and how a Large Language Model (LLM) was integrated to generate detailed recommendations for patients. Finally, it outlines the overall project plan and task allocation followed during the work.

3.1 Methodology

3.1.1 Overview

In this work, we develop a two-stage framework for peptic ulcer disease detection and recommendation generation using real upper GI endoscopy reports. The first stage focuses on structured prediction using classical machine learning. A cleaned and preprocessed dataset of 1,269 endoscopy samples is converted into numerical representations using TF-IDF features derived from textual fields such as Oesophagus, Stomach, Duodenum, Comments and Advice. Separate Logistic Regression models are trained to predict (i) the final diagnostic comment (six classes, e.g., Bleeding peptic ulcer, Normal upper GI findings, Gastritis with early ulcer formation) and (ii) the advised management plan (five classes, e.g., Medication, Surgery advised, Follow-up).

In the second stage, the predictions of these baseline classifiers are combined with patient-level information (age, sex and endoscopic findings) and passed as structured context to a Large Language Model (Gemini). The LLM synthesizes this information and produces a natural-language explanation and care recommendation for each case, emphasizing that its output is decision support and cannot replace a medical professional. Model performance is evaluated using standard metrics (accuracy, precision, recall, F1-score) and confusion matrices, while the LLM outputs are assessed qualitatively through example cases.

3.1.2 Proposed Methodology

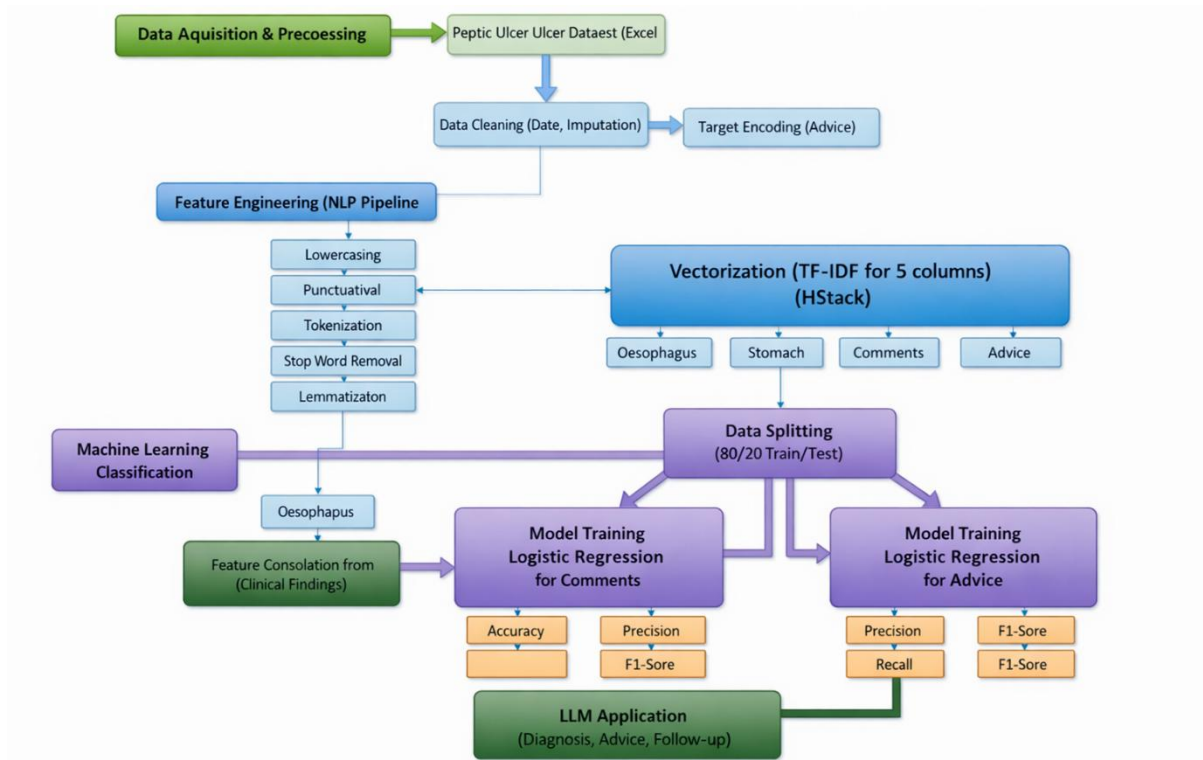


Figure 3.1: Methodology Diagram

The methodology diagram presents an end-to-end pipeline for the Peptic Ulcer Diagnosis & Advice System. First, patient data is collected from the Excel Peptic Ulcer dataset, then cleaned by handling missing values (imputation) and parsing the visit date, while the target output (advice) is encoded for learning. Next, an NLP feature-engineering pipeline preprocesses the clinical text (lowercasing, punctuation removal, tokenization, stop-word removal, and lemmatization). The processed text from key columns (e.g., oesophagus, stomach, comments, advice, etc.) is converted into numerical form using TF-IDF vectorization, and the features are horizontally stacked into a single feature set. After an 80/20 train-test split, two Logistic Regression classifiers are trained separately to predict Comments (diagnosis) and Advice, evaluated using metrics like accuracy, precision, recall, and F1-score. Finally, the predicted outputs are passed to an LLM layer to generate patient-friendly explanations, recommendations, and follow-up guidance.

3.1.3 Functional and Non-Functional Requirements

Functional Requirements:

1. Load the Excel dataset and validate required columns.
2. Select a patient (by ID/random) and show findings + basic info.
3. Clean data.
4. Preprocess text and extract TF-IDF features from findings.
5. Predict diagnosis/advice and show LLM-based simple explanation in the UI.

Non-Functional Requirements:

1. Fast response time for prediction and UI output.
2. Reliable operation with proper error handling (no crashes on bad/empty input).
3. Secure access and protect patient data privacy.
4. Maintainable and modular code for easy updates/debugging.
5. Accurate and consistent results with reproducible training setup.

3.1.4 Data Flow Diagram

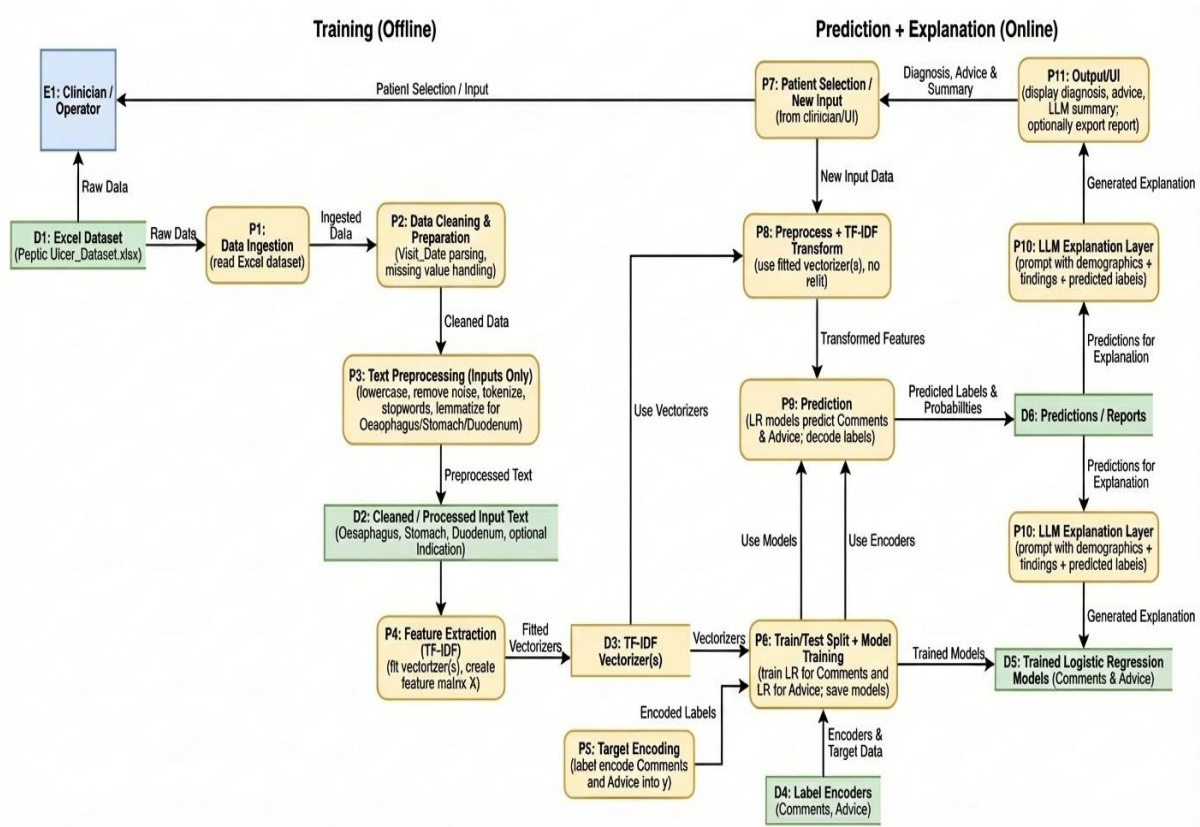


Figure 3.2: Data Flow Diagram

This data flow diagram illustrates the offline training and online prediction + explanation workflow of the Peptic Ulcer Diagnosis & Advice System. In the offline phase, patient records are loaded from an Excel dataset, cleaned, and text from endoscopic findings (oesophagus, stomach, duodenum) is preprocessed and converted into TF-IDF features. Labels for Comments and Advice are encoded, and Logistic Regression models are trained and saved along with the fitted TF-IDF vectorizers and encoders. In the online phase, a clinician inputs/selects a patient, the same TF-IDF transformation is applied using the saved vectorizers, and the trained models predict the diagnosis comment and advice. Finally, an LLM generates a simple explanation based on demographics, findings, and predicted labels, and the UI displays the final diagnosis, advice, and summary.

3.1.5 UI Design

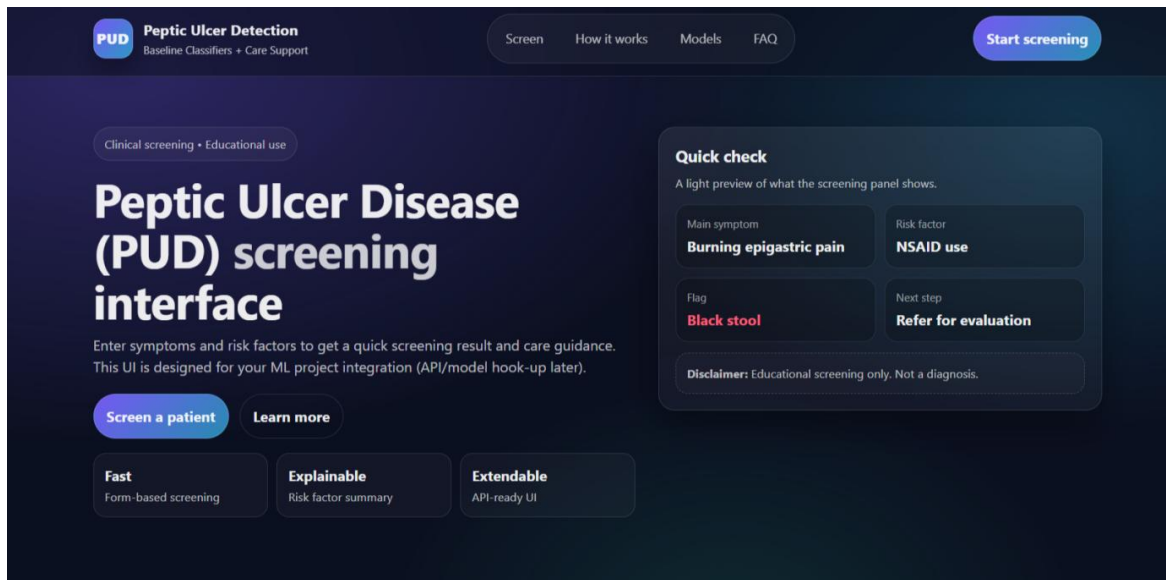


Figure 3.3: Landing Page

Figure 3.3 shows the entry point of the Peptic Ulcer Disease (PUD) screening system, where the user starts the clinical screening workflow. From here, the system provides a quick pre-check summary by mapping the provided symptom and risk-factor cues into an initial screening snapshot (main symptom, key risk factor, any warning flag, and the suggested next step). This screen explains the purpose of the system and guides the user to begin screening so the system can later generate a structured risk-oriented output and care guidance.

PUD Peptic Ulcer Detection
Baseline Classifiers + Care Support

Screen How it works Models FAQ [Start screening](#)

Patient ID (optional) e.g., DIU-001	Age e.g., 35
Sex Select	Epigastric pain (upper abdomen) Select
Pain timing Select (optional)	Heartburn / acid reflux Select (optional)
Nausea Select (optional)	Vomiting Select (optional)
Bloating / fullness Select (optional)	Unintentional weight loss Select (optional)
Black/tarry stool (melena) Select (optional)	Vomiting blood Select (optional)
NSAID use (e.g., ibuprofen) Select (optional)	Smoking Select (optional)
Alcohol Select (optional)	H. pylori test result Select (optional)

[Predict risk](#) [Reset](#)

Result Waiting

Predicted class —

Risk score —

Suggested care support (template)

- If severe pain or bleeding signs: seek urgent medical evaluation.
- Avoid NSAIDs unless clinician advises; consider gastro-protection.
- Discuss H. pylori testing/treatment options with a healthcare provider.
- Diet: smaller meals; reduce irritants (spicy, acidic) if they worsen symptoms.

Emergency signs: black stool, vomiting blood, fainting, severe abdominal pain. Contact emergency services immediately.

Figure 3.4: Screening Form

Figure 3.2 presents the main screening form and result generation stage of the system. The clinician enters patient information and selects relevant symptoms, risk factors, and test-related fields; the system then processes these inputs to produce the screening result panel. The output includes the predicted class and risk score, followed by a template-based care support section that provides recommended actions. If critical danger signs (such as black/tarry stool or vomiting blood) are selected, the system emphasizes urgent referral guidance to support safe and consistent triage decisions

3.2 Detailed Methodology and Design

3.2.1 Dataset

For this research, we use a real-world Peptic Ulcer Dataset collected from upper GI endoscopy procedures. The dataset contains 1,269 patient records and 13 features. The class distribution for Comments is relatively balanced: bleeding peptic ulcer (232 cases), peptic ulcer with gastric outlet obstruction (227), normal findings (220), GERD with ulcer disease (204), peptic ulcer disease (196) and gastritis with early ulcer formation (190). Similarly, Advice is distributed across follow-up, medication, diet modification, hemostasis done and surgery advised, each between ~239 and 274 cases. This balanced design is suitable for training baseline multi-class classifiers.

3.2.2 Data Cleaning and Preprocessing

Before building any model, several cleaning steps are applied to ensure consistency:

- i. **Date conversion:** the Visit Date column, originally stored as day-month-year strings, is converted to a proper datetime format so that temporal analysis or future longitudinal studies are possible.
- ii. **Missing value handling:** the Indication column contains 192 missing values. Instead of deleting these records, missing indications are imputed with the placeholder string 'Unknown'. This keeps all 1,269 cases while still distinguishing unknown indications from documented ones.
- iii. **Integrity check:** data types, ranges and unique counts are inspected to confirm there are no duplicate Patient_IDs or impossible values.

These preprocessing steps guarantee a clean tabular base over which text processing and model training can be performed reliably.

3.2.3 Text Normalization and TF-IDF Feature Extraction

We convert the free-text endoscopy fields into numerical features in three main steps:

- i. **Text Preprocessing:** Using NLTK, all five text columns (Oesophagus, Stomach, Duodenum, Comments, Advice) are cleaned with the same pipeline. We first convert text to lowercase and remove non-alphabetic characters. The text is then tokenized into words, common English stop-words (e.g. “and”, “the”, “with”) are removed, and the remaining words are lemmatized with WordNet so that inflected forms (e.g. “ulcers”, “erosions”) become their base forms (“ulcer”, “erosion”). Finally, the tokens are joined back into a cleaned sentence and stored in new “_processed” columns.
- ii. **TF-IDF features:** For each processed column, we train a separate TfidfVectorizer (maximum 1,000 terms per column). This converts every patient’s text into a sparse TF-IDF vector that reflects how important each term is within that field. The five resulting sparse matrices are horizontally

stacked into a single matrix X_{tfidf} , where each row represents a patient and each column represents a term from any of the text fields. This high-dimensional representation allows the model to capture patterns such as specific ulcer descriptions and their modifiers.

- iii. TF-IDF features: For each processed column, we train a separate `TfidfVectorizer` (maximum 1,000 terms per column). This converts every patient's text into a sparse TF-IDF vector that reflects how important each term is within that field. The five resulting sparse matrices are horizontally stacked into a single matrix X_{tfidf} , where each row represents a patient and each column represents a term from any of the text fields. This high-dimensional representation allows the model to capture patterns such as specific ulcer descriptions and their modifiers.

3.2.4 Target Label Encoding

The model predicts two targets: a diagnostic comment (with six possible classes) and management advice (with five possible classes). Because scikit-learn classifiers require numeric labels, each target is encoded separately using its own `LabelEncoder` instance. The “`label_encoder_comments`” converts each distinct comment string into an integer ID, while the “`label_encoder_advice`” performs the same mapping for each advice category. These encoded columns (“`Comments_encoded`” and “`Advice_encoded`”) are used as the y labels during model training, and the same encoders are later applied to translate the model's predicted integers back into readable text labels when generating recommendations.

3.2.5 Baseline Classification Models

Several classical models (Logistic Regression, Support Vector Machines, Random Forest) were explored, but the final experiments focused on Logistic Regression as a strong and interpretable baseline for text classification. For both prediction tasks—diagnostic comment and management advice, the same setup is used. The input is the combined TF-IDF feature matrix X_{tfidf} , while the outputs are the encoded labels: `Comments_encoded` for the diagnosis model and `Advice_encoded` for the advice model. Each task uses a separate `LogisticRegression` classifier with `max_iter = 1000` to ensure convergence and `random_state = 42` for reproducibility. The data is split using an 80/20 train-test split with the same random seed for both tasks. This results in two independent models: `logistic_model_comments`, which predicts the diagnostic comment class, and `logistic_model_advice`, which predicts the advice class. Model performance on the test sets is assessed using scikit-learn's classification report, providing precision, recall and F1-score per class, as well as macro and weighted averages. To make these results easier to interpret, helper functions are used to plot bar charts of precision, recall and F1-score for each class.

3.2.6 Evaluation Strategy

In addition to the classification reports, confusion matrices are created for both prediction tasks. For the Comments model, the confusion matrix shows how often one diagnosis is misclassified as another (for example, “bleeding peptic ulcer” predicted as “peptic ulcer disease”). For the Advice model, it highlights mix-ups between different recommendations such as Medication and Diet modification. By visualizing these errors, the confusion matrices reveal which classes are harder for the model to separate and help guide future improvements, such as re-balancing the data, adding new features, or trying more advanced models.

3.2.7 LLM-Assisted Recommendation Generation

The final part of the pipeline integrates a Large Language Model to turn model outputs into richer, patient-friendly explanations and care suggestions, without changing the classifier decisions themselves. For a selected patient (either randomly chosen or specified by Patient_ID), a small patient_data dictionary is created containing age, sex and the original findings for the oesophagus, stomach and duodenum. The same preprocess_text function and TF-IDF vectorizers used in training are then applied to these findings so that the new case is represented in the same feature space. The trained Logistic Regression models predict the most likely diagnostic comment and management advice, which are converted back to text labels using the fitted label encoders.

A structured prompt is then built that includes the patient’s demographics, endoscopic findings and the predicted diagnosis and advice. The prompt instructs the LLM to summarize the diagnosis in simple language, give concrete recommendations aligned with the predicted advice, suggest lifestyle changes and precautions, and clearly state that this is not a medical prescription and that the patient must consult a qualified doctor. Using the google.generativeai library, this prompt is sent to a Gemini model (e.g. gemini-flash-latest), and the returned text is treated as an LLM-generated care note. For each selected patient, the structured model predictions and the narrative LLM suggestions are shown side by side for interpretation and analysis

3.3 Project Plan

The project was carried out in several sequential phases. First, data understanding and cleaning were performed by loading the Excel dataset, inspecting the columns, converting date fields and handling missing values. Next, a text preprocessing pipeline was designed and tested using NLTK, including lowercasing, tokenization, stop-word removal and lemmatization on sample reports. Based on this, feature engineering was implemented by applying TF-IDF vectorization to multiple text fields and creating additional binary flags for key conditions such as ulcers, erosions and polyps. A baseline modelling phase followed, where Logistic Regression models were trained and evaluated for both diagnosis and advice prediction, along with

generating classification reports and confusion matrices. After establishing this baseline, LLM integration was added by connecting to the Gemini API, designing suitable prompts and implementing the `generate_llm_suggestions` function to combine model predictions with narrative recommendations. Finally, result analysis and documentation were conducted, including interpreting quantitative metrics, reviewing sample LLM outputs and preparing figures, tables and explanatory text for the thesis. This structured progression ensured that each component—data, models and LLM—was validated before moving to the next stage.

3.4 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 1 to week 27.

Table 3.1: Task Allocation and Timeline.

Tasks	Weeks														
	1	3	5	7	9	11	13	15	17	19	21	23	25	27	
Data collection															
Preprocessing all data															
Apply prediction model															
Apply LLM															
Performance analysis															

3.5 Summary

This chapter presented the complete research methodology of our work on peptic ulcer disease detection and care support. Starting from a real endoscopy dataset, we cleaned and normalized the data, transformed multi-field textual information into TF-IDF features and trained baseline Logistic Regression models for diagnosis and advice prediction. Additional binary features were derived to capture the presence of ulcers, erosions and polyps. On top of these structured predictions, we integrated a Large Language Model that generates detailed, human-readable recommendations for each case by combining patient information with model outputs. The next chapter discusses the experimental setup, quantitative results and example LLM outputs obtained from this methodology.

Chapter 4

Implementation and Results

This chapter presents the detailed implementation, evaluation, and results of the proposed peptic-ulcer decision-support system. We focus on the baseline machine-learning classifiers used for diagnosis and advice prediction, and on the Large Language Model (LLM) that generates patient-friendly recommendations on top of those predictions.

4.1 Environment Setup

The system was implemented in Python using a Jupyter Notebook environment. Most experiments were run on Google Colab so that GPU/CPU resources and all required libraries could be accessed easily without complex local configuration. The implementation relies on a small set of widely used open-source libraries. Table 4.1 lists the main tools and their roles.

Table 4.1: Used Libraries and Purpose

Library/ Tools	Purpose
Pandas, NumPy	Reading the Excel dataset, handling tables and numerical arrays.
scikit-learn	Text vectorization (TfidfVectorizer), label encoding, train-test splitting, Logistic Regression models, and evaluation metrics.
SciPy	Efficient storage and horizontal stacking of TF-IDF feature matrices.
NLTK / regular expressions	Tokenization, stop-word removal, lemmatization and cleaning of medical text.
Matplotlib, Seaborn	Plotting confusion matrices and simple bar charts for performance visualization.
Google.generativeai	Accessing the Gemini LLM API to generate narrative diagnosis summaries and care recommendations.

All experiments used the same random seed to keep results reproducible. The notebook was organized so that data loading, preprocessing, model training and LLM interaction can be executed step by step.

4.2 Testing and Evaluation

In this section we briefly describe how the proposed system was tested and how its performance was measured.

4.2.1 Data Preparation for Testing

After loading the Peptic Ulcer_Dataset into pandas, missing values in the Indication column were replaced with “Unknown” and the visit date was converted to proper datetime format. The main text fields (Oesophagus, Stomach, Duodenum, Comments, Advice) were cleaned using the same preprocess_text() function (lowercasing, removing non-letters, stop-word removal and lemmatization).

For each processed column a TfidfVectorizer was fitted and transformed into a sparse matrix. All TF-IDF matrices were then horizontally stacked into one combined feature matrix X. The Comments and Advice columns were encoded into integer labels using LabelEncoder, producing Comments_encoded and Advice_encoded. Finally, the data was split into training and test sets (80%–20%) using train_test_split with a fixed random seed.

4.2.2 Baseline Classifiers

Two Logistic Regression models (with max_iter = 1000) were trained on the training portion of X:

- i. one model for predicting the diagnostic comment class.
- ii. one model for predicting the advice class.

Both models were then applied to the respective test sets. Performance was evaluated using classification_report (accuracy, precision, recall, F1-score) and confusion matrices. The numerical results are summarized later in Table 4.2 and Table 4.3, and the confusion matrices are shown in Figure 4.1 and Figure 4.2.

4.2.3 LLM-Based Recommendation Check

For the LLM part, a small set of representative test cases was selected. For each case, the trained models predicted the comment and advice classes from the processed text. These predictions, together with age, sex and original findings, were used to build a structured prompt for the Gemini model. The generated recommendations were then inspected manually to see whether they matched the predicted labels, used simple language and respected the safety disclaimer. This qualitative check confirms that the LLM module can turn structured outputs into readable explanations for patients and clinicians.

4.3 Results and Discussion

In this section we briefly present the main results obtained from the baseline classifiers and the LLM module, and we discuss what they mean for our proposed system.

4.3.1 Comment Prediction Model

Table 4.2: Classification Report for Comments Prediction

Class	Precision	Recall	F1-score
Bleeding peptic ulcer	1.00	1.00	1.00
Gerd with ulcer disease	1.00	1.00	1.00
Gastritis with early ulcer formation	1.00	1.00	1.00

Normal upper GI findings	1.00	1.00	1.00
Peptic ulcer disease	1.00	1.00	1.00
Peptic ulcer with gastric outlet obstruction	1.00	1.00	1.00

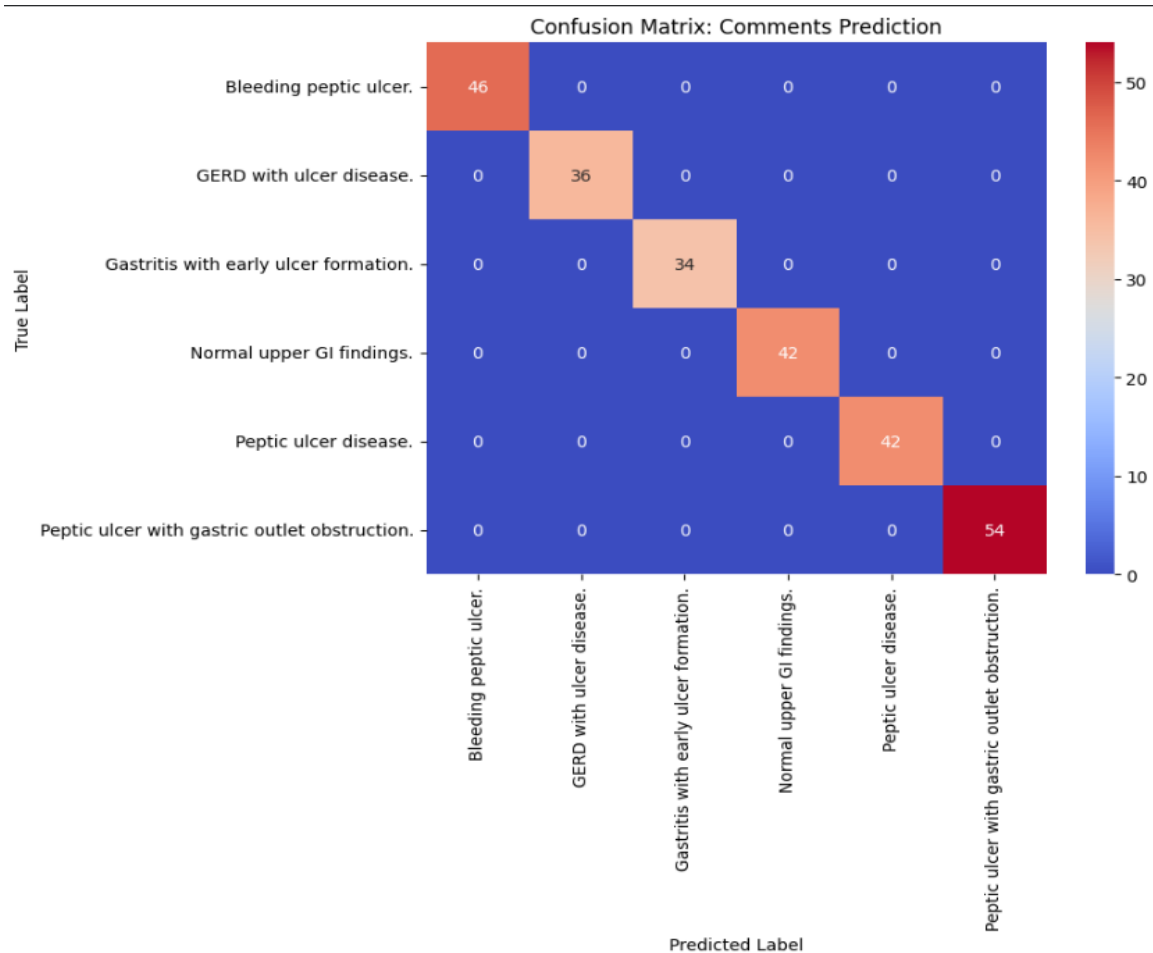


Figure 4.1: Confusion Matrix of comment prediction

The Logistic Regression model for diagnostic comment prediction performed very well on the test data. Most of the test samples were correctly classified into their respective comment classes. The confusion matrix (Figure 4.1) shows a clear diagonal pattern, which means that the predicted and true labels matched for almost all cases. Only a few errors appeared between clinically similar categories, such as “peptic ulcer disease” and “gastritis with early ulcer formation”, where the wording in the reports was very close. Overall, the results indicate that the TF-IDF features and baseline model are able to capture the main diagnostic patterns in the endoscopy text. The detailed class-wise metrics are summarized in Table 4.2.

4.3.2 Advice Prediction Model

Table 4.3: Classification Report for Advice Prediction

Class	Precision	Recall	F1-score
Diet modification	1.00	1.00	1.00
Follow-up	1.00	1.00	1.00
Hemostasis done	1.00	1.00	1.00
Medication	1.00	1.00	1.00
Surgery advised	1.00	1.00	1.00

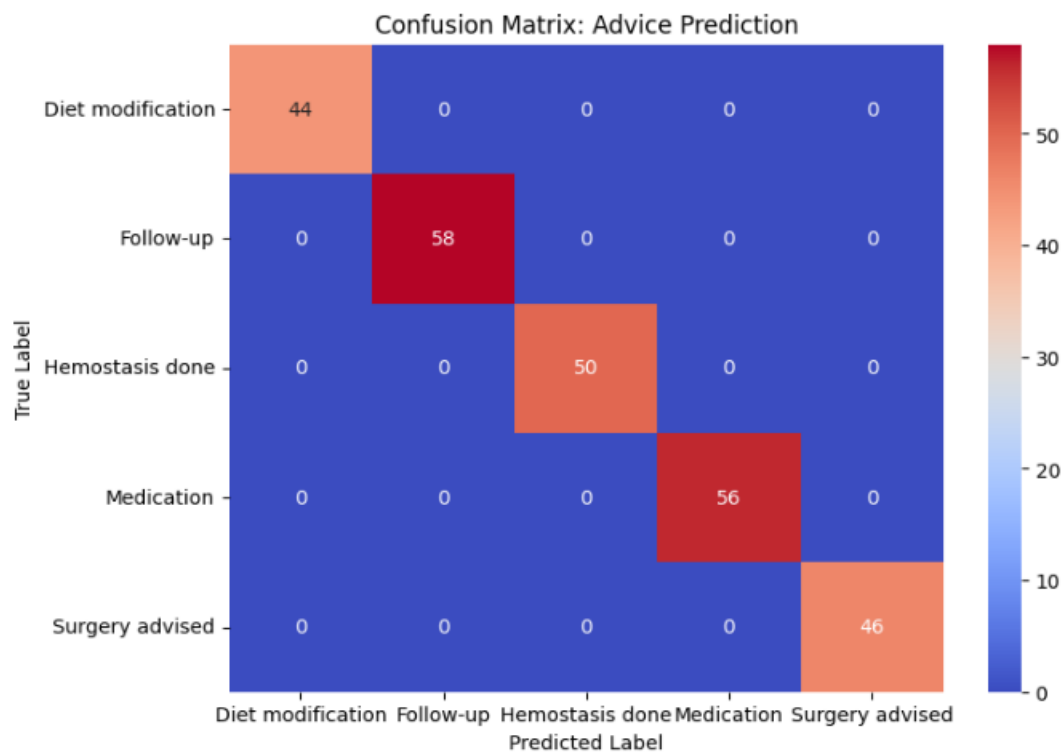


Figure 4.2: Confusion matrix of Advice prediction

The Logistic Regression model for advice prediction also achieved high performance. It correctly identified most of the test samples as Medication, Follow-up, Diet modification, Hemostasis done or Surgery advised. The confusion matrix (Figure 4.2) again shows that almost all points lie on the diagonal, with only minor confusion between conservative advice types such as medication and diet/lifestyle modification. This suggests that the language used in the findings and comments is strongly related to the final advice given by the endoscopist. The corresponding evaluation values for each advice class are listed in Table 4.3.

4.3.3 LLM-Based Recommendation Module

For the LLM module, the selected test cases showed that Gemini could generate clear and readable explanations based on the predicted comment and advice. For normal cases it

produced reassuring messages and simple lifestyle suggestions, while for ulcer and bleeding cases it discussed the importance of medication, follow-up and, when relevant, surgical consultation. In all prompts we included a safety note, and the LLM kept this tone by reminding us that the text is not a replacement for a real doctor. From these examples we can say that the combination of baseline classifiers and LLM is able to turn structured model outputs into patient-friendly narrative recommendations, which supports the main goal of this project.

4.4 Summary

In this chapter we described how the proposed peptic-ulcer decision-support system was implemented and evaluated. Using Python, common machine-learning libraries and the Peptic Ulcer_Dataset, we built TF-IDF feature representations of endoscopy reports and trained baseline Logistic Regression models to predict diagnostic comments and advice classes. Their performance was measured with standard multi-class metrics and visualized through confusion matrices.

On top of these structured predictions, a Gemini LLM was integrated to generate clear, patient-friendly explanations and care recommendations that remain aligned with the model outputs and include safety disclaimers. The results show that simple baseline classifiers, when combined with an LLM, can already provide a useful foundation for peptic-ulcer detection and care support.

Chapter 5

Engineering Standards and Design Challenges

This chapter explains the standards followed while developing the Peptic Ulcer Clinical Decision Support System and the challenges faced during its design. It describes how the system meets software, hardware, and communication standards, its effects on society and the environment, and its long-term sustainability. It also outlines the major technical problems and how they were solved.

5.1 Compliance with the Standards

The software development for this project followed widely accepted industry and academic standards to ensure reproducibility, compatibility, and maintainability.

5.1.1 Software Standards

The software development for this project followed widely accepted industry and academic standards to ensure reproducibility, compatibility and maintainability. Python 3.11 was used as the main programming language because of its popularity in data science and medical machine learning research and its strong ecosystem for text processing and modelling. Core libraries included pandas and NumPy for loading and manipulating the Excel dataset (“Peptic Ulcer_Dataset (1).xlsx”), Scikit-learn for TF-IDF vectorization, train-test splitting, label encoding and Logistic Regression models used to predict diagnostic comments and advice, and NLTK or similar text-processing tools for tokenization, stop-word removal and lemmatization of the oesophagus, stomach, duodenum, comment and advice fields. Matplotlib and related visualization libraries were used to generate accuracy curves, confusion matrices and other evaluation plots. In addition, a standard Large Language Model (LLM) API client was used to transform the model outputs into patient-friendly diagnostic summaries and recommendations. The peptic ulcer dataset was stored in an Excel (.xlsx) format with clearly defined columns such as Patient_ID, age, sex, indication, findings and outcomes. After preprocessing, cleaned datasets and TF-IDF feature matrices were saved in CSV format, while trained Logistic Regression models and label encoders were preserved in serialized files (such as .pkl or .joblib) to support future reuse. Evaluation artefacts, including confusion matrices, curves and reports, were exported as image and text files for platform-independent sharing and documentation.

5.1.2 Hardware Standards

The project was developed and tested using Google Colab as the primary development platform, which provided reliable CPU and optional GPU resources in a notebook environment and ensured easy access from different locations. In addition to the cloud

platform, a local desktop or laptop computer with a multi-core CPU and at least 8–16 GB of RAM was used to verify that the full pipeline, including preprocessing, TF–IDF feature extraction, Logistic Regression training and inference, and LLM prompt preparation, could run on standard hardware available to students and small clinics. The core machine learning workflow is mainly CPU-based, which keeps the system lightweight; optional GPU acceleration available in Google Colab was used only to speed up experimentation on larger feature spaces and batch processing. The LLM component runs through an external cloud API, so no additional local GPU resources are required for generating explanations. For storage, Solid State Drives (SSD) were used on the local system to ensure fast loading of the Excel dataset, intermediate CSV files and model artefacts, thereby reducing I/O bottlenecks during preprocessing and evaluation. Google Drive was used as an online backup location for notebooks, models and result figures, supporting both data safety and collaboration.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The proposed system enables fast and consistent analysis of upper gastrointestinal (GI) endoscopy findings related to peptic ulcer and other common conditions. By processing the oesophagus, stomach and duodenum text fields and predicting standardized diagnostic comments and advice, the system can support gastroenterologists and general physicians in making quicker and more uniform decisions. It can help identify high-risk patients earlier, so that peptic ulcer and related complications are treated in time, and it provides a structured way to avoid missing important findings in long free-text reports. Through the LLM component, the system can also convert these technical outputs into clear, patient-friendly explanations and lifestyle suggestions, which improves patient understanding and increases adherence to treatment and follow-up recommendations. This is particularly valuable in settings where experienced specialists are not always available, as the system acts as a decision support tool that enhances the overall quality of care.

5.2.2 Impact on Society & Environment

From a societal perspective, the system reduces the dependence on highly specialized gastroenterologists for the initial interpretation of endoscopy reports by offering a standardized second opinion based on historical data. Junior doctors and medical officers can use the model's predictions as guidance when drafting reports and planning treatment, which leads to more consistent documentation and better triage of patients. Severe or suspicious cases can be flagged more systematically for specialist review, while routine or low-risk cases can be managed using clear advice generated by the system. This increases access to quality GI care in smaller hospitals and rural or semi-urban clinics, where specialist availability is limited. From an environmental perspective, the system contributes indirectly to resource conservation. By improving diagnostic accuracy and standardizing advice, it can reduce unnecessary repeat endoscopies, avoid redundant hospital visits and limit overuse of medications that may be prescribed because of misinterpretation or incomplete documentation. Fewer avoidable procedures and visits mean lower consumption of energy, consumables and transport resources, which in turn helps reduce the environmental footprint of healthcare services.

5.2.3 Ethical Aspects

Ethical considerations played an important role throughout the development of the system. The peptic ulcer dataset used in this project was anonymized before analysis, and

no direct personal identifiers such as names, phone numbers or addresses were included in the modelling process. Only non-sensitive attributes like age, sex, indication and textual findings were used. The system follows key ethical AI principles: patient privacy is protected by design; the preprocessing steps, TF-IDF settings and model parameters are documented to support transparency; and the outputs of the Logistic Regression and the LLM are intended as decision support, not as a replacement for clinical judgment. Final diagnostic and treatment decisions remain under the responsibility of qualified medical professionals. Furthermore, the project acknowledges that the dataset originates from a specific clinical environment, so the model may reflect local patterns. Any deployment to new hospitals or populations must therefore be accompanied by careful re-evaluation of performance across different age groups, genders and indication patterns to detect and mitigate potential biases.

5.2.4 Sustainability Plan

The sustainability plan for the peptic ulcer decision support system is based on low resource requirements, reliance on open-source tools and an easily reproducible workflow. As new endoscopy cases are collected, the underlying dataset can be expanded and the same preprocessing, TF-IDF and Logistic Regression training scripts can be rerun to update or fine-tune the models without redesigning the system. Because all steps, from reading the Excel file to saving evaluation plots, are implemented in Python notebooks, future students or engineers can reproduce the experiments, introduce modifications and validate improvements. The trained models are lightweight enough to be deployed as a simple web application on a modest server or cloud instance, which keeps deployment and maintenance costs low. The LLM integration is accessed via an external API, so improvements in language model technology can be adopted without altering the core classifier. Finally, the framework can be shared with hospitals and universities to test on different patient populations, incorporate additional GI conditions and refine performance, ensuring that the system remains clinically relevant, technically robust and sustainable over the long term.

5.3 Project Management and Financial Analysis

The project was executed under budget-conscious planning, ensuring efficient resource utilization. A detailed financial breakdown is provided below:

Table 5.1: Budgets of project

Components	Estimated Cost (BDT)
Internet and Cloud Subscription	1,100 – 2,600
Software Licenses	1,500
Data Collection	1,500
Others	1,900 – 2,400
Total Estimated Cost	6,000 - 8,000 BDT

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories.

Table 5.2: Mapping with Complex Engineering Problem.

EP1 Dept of Knowled ge	EP2 Range Of Conflicting Requireme nts	EP3 Depth of Analys is	EP4 Familiari ty of Issues	EP5 Extent of Applica ble Codes	EP6 Extent Of Stake- holder Involveme nt	EP7 Interdepende nce
✓		✓	✓			✓

EP1: Required advanced understanding of clinical text processing, TF-IDF feature extraction, Logistic Regression behavior, and basic medical knowledge about peptic ulcer and other upper GI conditions. In addition, prompt design for the LLM was needed so that it could correctly interpret the model outputs and generate safe, clear explanations.

EP3: This was a complex task because we had to deeply understand noisy, unstructured endoscopy text, design a robust preprocessing pipeline, select appropriate parameters for TF-IDF and Logistic Regression, and make sure the predictions aligned with actual medical comments and advice.

EP4: Some challenges, like handling missing values, cleaning text, dealing with class imbalance and evaluating models, were familiar to data engineers. However, mapping free-text medical findings into structured labels and then generating clinically meaningful natural language summaries was less familiar and required careful analysis and iteration.

EP7: Different tasks of the system—data cleaning, text preprocessing, TF-IDF vectorization, label encoding, model training, evaluation and LLM prompt construction—were interconnected. A small change in preprocessing (such as modifying stop-word lists or n-gram ranges) could change the feature space, affect model performance, and finally alter the explanations produced by the LLM.

Mapping with Knowledge Profile

This section is designed to map the overall problem and EP1 (*multiple between K3, K4, K5, K6, K8 for attaining EP1*) to the Knowledge Profile.

Table 5.3: Mapping with knowledge Profile.

K1 Natu ral Scien ce	K2 Mathem atics	K3 Engineeri ng Fundame ntals	K4 Special ist Knowle dge	K5 Enginee ring Design	K6 Enginee ring Practice	K7 Comprehe nsion	K8 Resear ch Literat ure
	✓	✓		✓	✓		✓

K2: Strong use of statistical concepts, linear algebra and probability was required in TF-IDF feature extraction, model training, evaluation metrics (accuracy, precision, recall, F1-score) and interpretation of confusion matrices.

K3: Engineering fundamentals of building an end-to-end information processing system were applied. This included acquiring data from Excel, designing a preprocessing pipeline, building the ML model, and integrating the outputs into a higher-level application (LLM explanations).

K5: The system was designed step by step to handle complex medical text data, ensuring it is efficient, reproducible and scalable. Design decisions were made about which text fields to use, how to represent them numerically, how to structure the diagnostic labels, and how to present the final outputs to users.

K6: Practical tools and platforms were used: Python, Scikit-learn, NLTK, Google Colab and standard APIs for calling the LLM. The knowledge was applied in a real-world setting by programming the complete pipeline, running experiments, debugging errors and improving model performance.

K8: Research literature on peptic ulcer, endoscopy reporting standards, clinical natural language processing, classical machine learning models and LLM-based medical summarization was reviewed. This helped select suitable methods, avoid known pitfalls and design a system aligned with current research trends.

5.4.2 Engineering Activities

Mapping with Complex Engineering Activities

Table 5.4: Mapping with Complex Engineering Activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓		✓	✓	✓

EA1: We needed a wide range of resources, including computing power for running TF-IDF and Logistic Regression, data storage for the peptic ulcer dataset and intermediate files, cloud services (Google Colab) for experiments, and software tools (Python, Scikit-learn, NLTK, LLM API).

EA3: The project is innovative because it combines a traditional machine learning classifier (Logistic Regression) with a Large Language Model. The ML model provides structured, reliable predictions of diagnostic comments and advice, while the LLM converts these results into simple language that patients and non-specialist doctors can understand.

EA4: This system can help doctors diagnose and manage peptic ulcer and related upper GI conditions early, improving patient outcomes. It also contributes to more efficient use

of hospital resources and can reduce unnecessary procedures and travel, which has positive consequences for society and the environment.

EA5: Some activities were familiar, such as cleaning tabular data, applying TF-IDF, and training classical ML models. However, working with real clinical text, designing medically safe prompts for the LLM, and thinking about deployment in a hospital context were more challenging and required learning new concepts, which made the project both demanding and enriching.

5.5 Summary

In this chapter, we explained why our peptic ulcer clinical decision support system is a complex engineering project. Building it required knowledge of medical text preprocessing, TF-IDF feature extraction, Logistic Regression modelling and Large Language Model-based explanation generation. We also had to connect several parts of the system—data cleaning, feature extraction, label encoding, model training, evaluation and text generation—so they work together smoothly. By mapping the work to complex engineering problem categories, knowledge areas and engineering activities, we showed that this project had to deal with different requirements, follow technical and ethical rules, and use new ideas in healthcare and artificial intelligence. All of this means the system needed skills from different fields and a careful, step-by-step engineering process to make it work well and to make it useful in real clinical settings for peptic ulcer management.

Chapter 6

Conclusion

This chapter highlights the main outcomes of our work on peptic ulcer disease detection and care support. It gives a short summary of what we have done, mentions the main limitations of the system, and suggests possible directions for future improvement.

6.1 Summary

In this work, we designed and implemented a peptic ulcer decision-support system based on real upper GI endoscopy reports. The system uses only the report information stored in an Excel file, including patient age, sex and the free-text findings for oesophagus, stomach and duodenum, together with the final Comment and Advice written by the endoscopist. We first cleaned the dataset, handled missing values and normalized the text using lowercasing, stop-word removal and lemmatization. Then we converted the main text fields into TF-IDF features and trained baseline Logistic Regression models to predict two targets: diagnostic comment class and advice class. The models were evaluated using standard metrics such as accuracy, precision, recall and F1-score. On top of these baseline classifiers, we integrated a Large Language Model (LLM). For a selected patient, the system combines their age, sex, endoscopic findings and the predicted comment and advice, and sends this information to the LLM with a structured prompt. The LLM then generates a short explanation of the condition and general care suggestions in simple language, always reminding that the output is only for support and that final decisions must be made by a doctor.

Overall, our work shows that even simple baseline models, when combined with LLMs, can form the basis of a text-based peptic ulcer decision-support tool that helps translate structured predictions into patient-friendly recommendations.

6.2 Limitation

Although the project achieved its basic objectives, there are some important limitations:

- i. The dataset comes from a single center and a limited time period. The model has not been validated on data from other hospitals or populations, so its generalization ability is unknown.
- ii. In the current implementation, TF-IDF features are built not only from the findings but also from the processed Comments and Advice text. This makes the classification task easier and can lead to over-optimistic performance. A more realistic setup would predict comment and advice only from findings and basic clinical information.

- iii. We used only baseline Logistic Regression models and did not compare many different algorithms or perform extensive hyper-parameter tuning. More advanced text models could perform differently.

6.3 Future Work

There are several directions in which this research can be extended in future:

- i. Remove label leakage and retrain the models using only oesophagus, stomach and duodenum findings (plus basic patient data) as input features, so that comment and advice are truly predicted rather than echoed.
- ii. Collect and test on multi-centre datasets from different hospitals and time periods to study how well the models generalize and to reduce bias.
- iii. Explore stronger text models, such as transformer-based classifiers or clinical word embeddings, and compare them with Logistic Regression on the same feature space.
- iv. Systematically evaluate the LLM outputs with clinicians, using checklists for medical correctness, completeness and readability, and refine the prompt design based on their feedback.

By addressing these points, the proposed system can move from a proof-of-concept prototype towards a more robust and clinically useful tool for peptic ulcer disease detection and care support.

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